

Weekly Progress Report

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Domain: Python Development Internship

Internship Duration: June – July 2026

Date of Submission: 12 July 2026

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1. OVERVIEW

This report summarizes my **4-week Python Development Internship** journey. During this internship, I built **four complete Python projects** from scratch: a **URL Shortener**, a **File Organizer**, a **Password Manager**, and a **Quiz Game**.

I started with basic scripts in Week 1, added advanced features (REST APIs, PostgreSQL, Timer, Encryption, Logging) in Week 2, prepared documentation and testing in Week 3, and finalized everything for submission in Week 4. This internship helped me develop practical skills in web development, database management, security, and software documentation.

2. WEEK 01 – PROJECT DEVELOPMENT

Focus: Building the Core Projects

Duration: 23 June 2026 – 29 June 2026

Achievements:

1. URL Shortener:

- Built a web application using Flask and SQLite.
- Implemented a CLI mode for terminal-based shortening.

- Generated unique short codes using SHA256 hashing.

2. File Organizer:

- Created a command-line utility to sort files by extension.
- Categorized files into Images, Documents, Videos, Audio, Archives, etc.
- Added a --dry-run option to preview changes.

3. Password Manager:

- Developed a secure password storage system with Master Password authentication.
- Used Fernet (symmetric encryption) to encrypt stored passwords.
- Added a built-in password generator.

4. Quiz Game:

- Designed an interactive quiz using JSON data storage.
- Supported Multiple Choice Questions (MCQ).
- Implemented a scoring system.

Lessons Learned (Week 01):

- Understanding Flask web framework basics.
- Working with SQLite databases.
- Basic encryption using the cryptography library.

3. WEEK 02 – ENHANCEMENTS & APIs

Focus: Adding Advanced Features & Making Projects Production-Ready

Duration: 30 June 2026 – 06 July 2026

Achievements:

1. URL Shortener (Enhanced):

- Created **REST APIs** (/api/shorten, /api/expand) for programmatic access.
- Added **API Key Authentication** using Flask-HTTPAuth.
- Added **Copy-to-Clipboard** button in the web UI.
- Added **Click Counter** to track link usage.
- Integrated **PostgreSQL** support for cloud deployment (Render.com).

2. Quiz Game (Enhanced):

- Added a **10-second Timer** using Python's threading module.
- Introduced **Difficulty Levels**: Easy (10 sec), Medium (7 sec), Hard (5 sec).

- Implemented **Bonus Points** for quick answers.
- Expanded the question bank to **30+ questions**.
- Added an **SQLite High Score Leaderboard**.

3. File Organizer (Enhanced):

- Refactored code to follow **PEP 8** standards.
- Added **Custom Category Support** via config.json.
- Implemented **Logging** – all file movements are recorded.
- Added **duplicate file handling** (auto-renaming).

4. Password Manager (Enhanced):

- Upgraded Master Password hashing from SHA256 to **bcrypt**.
- Added a **Password Strength Checker** (Weak/Medium/Strong).
- Implemented **Export/Import** functionality to JSON.
- Fixed Git security issues with .gitignore.

Challenges Faced:

- **Flask-HTTPAuth Installation:** Resolved by running pip install flask-httpauth.
- **Timer Implementation:** Learned threading to create a non-blocking timer.
- **PostgreSQL Connection Error:** Fixed the URL format (postgres:// to postgresql://).

Lessons Learned (Week 02):

- REST API development using Flask.
- Threading and concurrency in Python.
- Importance of environment variables for deployment.

4. WEEK 03 – DOCUMENTATION & TESTING

Focus: Testing, Documentation, and Portfolio Development

Duration: 07 July 2026 – 13 July 2026

Achievements:

1. User Manuals & Documentation:

- Created detailed **User Manuals** for all 4 projects.
- Added installation steps, required libraries, and execution processes.
- Included screenshots for better understanding.

2. Unit Testing:

- Learned **pytest** and wrote test cases for critical modules.
- Verified expected outputs and reduced runtime bugs.
- Tested edge cases (invalid URLs, empty folders, wrong passwords).

3. Portfolio Website:

- Started building a portfolio website using HTML, CSS, and JavaScript.
- Added project descriptions, screenshots, skills, and GitHub links.
- Made the website responsive for desktop and mobile.

4. GitHub Management:

- Organized all project files into separate folders.
- Created README files for every project.
- Uploaded all code to a public GitHub repository.

Lessons Learned (Week 03):

- Importance of unit testing for bug prevention.
- Creating professional documentation.
- Frontend basics for portfolio websites.

5. WEEK 04 – FINAL SUBMISSION

Focus: Final Testing, Bug Fixing, and Project Submission

Duration: 12 July 2026

Achievements:

1. Final Code Optimization:

- Fixed minor bugs in all 4 projects.
- Improved UI/UX for the URL Shortener.
- Verified database handling and error messages.

2. Final Testing:

- Performed functional, validation, and performance testing on all projects.
- Tested the URL Shortener on different browsers.
- Verified export/import functionality in the Password Manager.

3. GitHub Live Session Recording:

- Recorded a complete walkthrough of all projects.
- Demonstrated how to run each project and explained the code.

- Uploaded the recording to the submission portal.

4. Final Submission:

- Uploaded all 4 weekly reports (Week 1-4 PDFs).
- Uploaded the final consolidated report (this document).
- Submitted all project source code via GitHub and ZIP file.